Adaptive Client Selection for Multimodal Federated Vision Foundation Models in Alzheimer's Disease Prediction

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Declaration

We do hereby declare that the research work presented in this report entitled, "Adaptive Client Selection for Multimodal Federated Vision Foundation Models in Alzheimer's Disease Prediction" is the result of our own work. We further declare that this report has been compiled and written by us, and no part of this report has been submitted elsewhere for the requirements of any degree, award, diploma, or any other purpose except for publications. The materials that are obtained from other sources are duly acknowledged in this report.

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Chapter 1

Introduction

Alzheimer's disease (AD) diagnosis in practice depends on two kinds of evidence that are usually held apart: brain-imaging scans (MRI or PET) that show structural change, and clinical or demographic records that capture cognitive test scores, genetic risk markers, and lifestyle history. A moderator of AD risk who reads only the scan misses the clinical context that determines how alarming a given degree of atrophy is, and a system that reads only the clinical record misses the structural evidence a scan provides. Both kinds of evidence are, in addition, collected by different hospitals and research centres, and regulatory restrictions on patient data mean this information cannot simply be pooled onto one server for training. This chapter states the resulting problem, the objectives adopted in response, and the contributions this project claims.

1.1 Problem Statement

Alzheimer's disease is a progressive neurodegenerative disorder whose early and accurate diagnosis remains a major clinical challenge, since reliable detection typically requires the joint analysis of neuroimaging data (MRI/PET) and clinical or demographic risk factors (age, genetic markers, cognitive scores) collected across multiple hospitals and research centres. Centralising this sensitive medical data to train a single large-scale model is often infeasible because of patient privacy regulations, data-sharing restrictions between institutions, and the heterogeneity of imaging protocols and patient populations across sites. Federated Learning (FL) offers a privacy-preserving alternative by allowing multiple clients (hospitals or sites) to collaboratively train a shared model without exchanging raw data [9]. However, existing federated frameworks for Alzheimer's diagnosis largely rely on naive or random client participation strategies, ignoring differences in data quality, modality completeness, and class distribution across clients [1, 3, 4]. This can lead to biased global models, slower convergence, and reduced diagnostic reliability, especially when pretrained vision foundation models are fine-tuned in such heterogeneous, non-IID federated settings. There is therefore a need for an adaptive client selection mechanism, integrated with a multimodal vision foundation model, that can identify and prioritise the most informative and reliable clients during federated training so as to improve the accuracy, fairness, and efficiency of Alzheimer's disease prediction.

1.2 Problem Background

Deep learning applied to neuroimaging has repeatedly shown that convolutional and transformer-based encoders can recover diagnostically useful structure from MRI and PET scans that is difficult to specify by hand [2]. Vision foundation models pretrained at scale on large image or vision-language corpora, such as the contrastive vision-language approach of Radford et al. [8], have extended this further by supplying representations that transfer to a new task with comparatively little labelled data, and recent medical adaptations such as ADLIP [2] apply this idea directly to Alzheimer's diagnosis by jointly encoding MRI scans and clinical text. These systems, however, are trained in a centralised setting, which is difficult to reconcile with the privacy constraints that govern hospital data.

Federated Learning [9] was proposed precisely to remove the need for centralisation: clients train locally and only model updates, not raw data, are exchanged with a central server. Mohanraj and Radhakrishnan [1] demonstrate that a federated, multimodal approach can reach competitive accuracy for Alzheimer's diagnosis while preserving privacy, and Mouheb et al. [3] show that a pretrained 3D medical foundation model (SAM-Med3D) can be fine-tuned under federated constraints, with the choice of aggregation strategy and encoder-freezing configuration measurably affecting performance under non-IID data. Parameter-efficient fine-tuning methods such as LoRA [7] make this style of federated fine-tuning practical by updating only a small low-rank subset of parameters, which reduces the size of the updates each client must transmit.

What remains largely unaddressed in this line of work is *how clients are chosen* to participate in a given round of training. The default in almost all of the systems above is random sampling or full participation. The broader federated learning literature shows that this default is not benign: Li, Chen, and Teng [4] survey a wide range of client selection strategies and show that client heterogeneity is a major factor in convergence speed and fairness, and Tang et al. [12] propose FedCor, which uses the correlation structure between client updates to select a subset expected to reduce the global loss fastest. Neither of these general-purpose strategies is designed for the specific constraints of multimodal medical data, where a client hospital may have an MRI scan without a matching clinical record, may serve a population skewed toward a particular disease stage, or may simply have lower image quality than another site. This project is positioned at the intersection of these two bodies of work: it takes the federated, multimodal, foundation-model pipeline that recent Alzheimer's-diagnosis work has established, and combines it with a client selection mechanism that is explicitly aware of the data-quality and modality constraints that arise in a real hospital network.

1.3 Research Objectives

- To design a multimodal federated learning framework that fuses neuroimaging (MRI) data with clinical and demographic risk factors for Alzheimer's disease prediction.
- To integrate a pretrained vision foundation model into the federated pipeline using parameter-efficient fine-tuning (LoRA) [7] so that the communication cost of each round is kept low enough for resource-constrained hospital clients.
- To develop and evaluate an adaptive client selection strategy that scores clients on local data quality, modality completeness, and class balance, and that selects the subset of clients most likely to improve the global model in each round.
- To compare the proposed client-selection-aware framework against standard baselines, including FedAvg [9] with random client sampling and correlation-based selection [12], on diagnostic accuracy, convergence speed, and communication efficiency.
- To make model decisions and client contributions interpretable to clinicians using Grad-CAM [10] for the imaging branch and SHAP [11] for the clinical branch.

1.4 Motivations

Three observations motivated this project. The first is regulatory and practical: hospital data cannot be centralised for training, so any system intended for real deployment across multiple sites has to be federated by design, not as an afterthought. The second observation concerns the data itself. A federation of hospital clients is rarely uniform: sites differ in scanner hardware and imaging protocol, in the demographic composition of their patient population, and in how completely they have recorded clinical variables alongside imaging. Treating every client as equally informative, as random or full-participation sampling does, discards information that is available almost for free once each client can report a quality or completeness score for its own data. The third observation concerns trust. A clinician who is asked to act on a federated model's prediction has no way to inspect the training process directly, and if the contribution of an unreliable client cannot be discounted, the resulting prediction is harder to trust than one from a system where client quality is explicitly accounted for and where the prediction itself can be explained through Grad-CAM and SHAP.

1.5 Significance of the Research

If successful, this project contributes a client selection mechanism that is reusable beyond Alzheimer's diagnosis, since the underlying problem, that federated clients vary in reliability and completeness, is common to most multi-hospital medical AI applications. The communication-efficiency component, built around LoRA fine-tuning of a foundation model, is significant for low-resource hospital settings that cannot afford to transmit full model updates every round. The explainability component addresses a practical adoption barrier: clinicians are considerably more likely to trust and act on a system whose predictions can be traced to specific imaging regions or clinical variables. Finally, by directly comparing an adaptive selection strategy against random sampling and against a general-purpose strategy such as FedCor [12], the project produces evidence on whether medical-domain-aware selection criteria offer a measurable benefit over domain-agnostic ones, which is a question the existing literature has not yet answered for this setting.

1.6 Research Contributions

- A federated learning framework in which a shared vision foundation model encoder and a clinical feature encoder are fused and fine-tuned locally using LoRA, so that only a small set of low-rank parameters is communicated in each round.
- An adaptive client selection algorithm that scores each client on local data quality, modality completeness, and class balance before every aggregation round, in place of the random or full-participation sampling used in prior federated Alzheimer's-diagnosis work [1, 3].
- A weighted aggregation scheme in which the contribution of each selected client is scaled by its selection score, extending standard FedAvg [9] rather than replacing it outright.
- An explainability layer that pairs Grad-CAM [10] attributions on the imaging branch with SHAP [11] attributions on the clinical branch, so that a flagged

prediction is accompanied by evidence from both modalities.

- A planned ablation study isolating the effect of client selection, of the weighted aggregation scheme, and of each modality in turn, to establish which components of the framework are responsible for any observed improvement over the FedAvg and FedCor baselines.

1.7 Complex Engineering Analysis

The project is assessed below against the attributes that characterise a complex engineering problem and the activities required to address it.

1.7.1 Complex Engineering Problems

Table 1.1: Mapping of the project onto complex engineering problem attributes.

Attribute	How the project exhibits it
WP1: Depth of knowledge	Requires an understanding of transformer-based vision foundation models, parameter-efficient fine-tuning, federated optimisation, and statistical measures of client heterogeneity, spanning several specialised subfields at once.
WP2: Range of conflicting requirements	Diagnostic accuracy, communication efficiency, patient privacy, and fairness across heterogeneous clients compete with one another; improving one can degrade another, e.g. selecting only the highest-quality clients raises accuracy but can reduce fairness toward under-represented sites.
WP3: Depth of analysis	Requires analysing how non-IID data distributions across clients affect convergence and bias in the aggregated model, and designing a scoring function that captures data quality, modality completeness, and class balance simultaneously.
WP4: Familiarity of issues	Client heterogeneity is a known challenge in general federated learning, but combining it with a multimodal vision foundation model for a specific clinical task is not a standard, well-trodden combination.
WP6: Extent of stakeholder involvement	Involves hospitals contributing data, clinicians who must trust and act on predictions, and, indirectly, the patients whose diagnosis depends on the system's reliability.
WP7: Interdependence	The client scoring module, the weighted aggregation scheme, and the foundation-model fine-tuning procedure are mutually dependent: changing the scoring function changes which clients are aggregated, which in turn changes what the shared encoder learns.

1.7.2 Complex Engineering Activities

Table 1.2: Mapping of the project onto complex engineering activity attributes.

Attribute	How the project exhibits it
EA1: Range of resources	Involves pretrained vision foundation model checkpoints, public MRI datasets (ADNI, OASIS-3), a federated learning framework, and GPU resources for repeated multi-client training runs.
EA2: Level of interaction	Requires resolving interactions between statistical heterogeneity across clients, the communication budget imposed by LoRA fine-tuning, and overall model performance.
EA3: Innovation	Introduces a data-quality-aware client selection mechanism combined with a multimodal vision foundation model for Alzheimer's diagnosis, a combination not present in the reviewed literature.
EA4: Consequences to society	Improves the reliability of early Alzheimer's diagnosis while preserving patient privacy across institutions, with direct benefit to healthcare systems operating under real data-governance constraints.
EA5: Familiarity	Extends standard federated learning practice into a combination, adaptive medical client selection under a foundation-model fine-tuning regime, that has no established precedent to follow directly.

1.8 Thesis/Project Organization

The remainder of this report is organised as follows. Chapter 2 reviews the literature on deep learning and vision foundation models for Alzheimer's diagnosis, federated learning for medical applications, client selection strategies in federated learning, and explainability in medical AI, and closes with an analysis of the gap this project addresses. Chapter 3 presents the research methodology: the proposed framework, the constituent datasets and their preparation, the model and algorithm design, and the planned explainability and ablation components. Chapter 4 will report the implementation and result analysis once experiments are conducted. Chapter 5 states the standards, constraints, and ethical considerations that apply to the work, together with the project timeline. Chapter 6 will conclude the report and identify directions for further work.

Summary

This chapter has stated the problem this project addresses: Alzheimer's diagnosis requires combining imaging and clinical evidence collected across institutions that cannot share raw data, and existing federated approaches to this problem select participating clients without regard to data quality or completeness. The project responds by proposing an adaptive, quality-aware client selection mechanism layered onto a multimodal, LoRA-fine-tuned vision foundation model. Chapter 2 surveys the relevant literature in detail and formalises the gap this project takes up.

Chapter 2

Background Study

This chapter surveys the work on which the project builds and establishes the gap it addresses. The literature is organised into four themes: deep learning and vision foundation models for Alzheimer's diagnosis, federated learning for medical applications, client selection strategies in federated learning, and explainability in medical AI. Section 2.2 draws these threads together into the problem this work takes up.

2.1 Literature Review

2.1.1 Deep Learning and Vision Foundation Models for Alzheimer's Disease Diagnosis

Vision foundation models pretrained on large corpora have changed how imaging tasks are approached, replacing task-specific architectures with a large pretrained encoder that is adapted to the target task. Radford et al. [8] introduced CLIP, a contrastive vision-language pretraining approach that aligns image and text representations in a shared embedding space, and this style of pretraining has since been adapted directly to medical imaging. Lin et al. [2] apply the same principle in ADLIP, a vision-language foundation model that jointly encodes MRI scans and structured clinical data for Alzheimer's diagnosis, reporting strong diagnostic performance and even zero-shot prediction ability. This confirms that a vision foundation model, adapted with clinical context, is an effective architecture for this task.

Gap. ADLIP and comparable systems are trained in a centralised setting. They do not address how a vision foundation model should be adapted when the imaging and clinical data it needs are distributed across institutions that cannot share raw data with one another.

2.1.2 Federated Learning for Medical Applications

Federated Learning was formalised by McMahan et al. [9], who proposed FedAvg, in which clients train locally and a central server averages their model updates, weighted by local dataset size, to produce a global model. This formulation underlies almost all subsequent federated medical AI work, including the two studies closest to this project. Mohanraj and Radhakrishnan [1] propose FuzzyFed-CNN, combining MRI-based CNN features with a fuzzy inference layer to produce explainable, federated predictions for early Alzheimer's diagnosis across several simulated hospital clients, and report that the approach can reach high accuracy while preserving privacy. Mouheb et al. [3] benchmark the federated fine-tuning of SAM-Med3D, a pretrained 3D medical foundation model, for dementia classification, comparing classification-head architectures, fine-tuning strategies, and aggregation methods, and find that freezing the foundation model's encoder can match full fine-tuning while advanced aggregation methods outperform plain FedAvg. Parameter-efficient fine-tuning, in particular LoRA [7], is what makes this style of federated foundation-model adaptation practical, since it updates only a small low-rank matrix pair per layer instead of the full parameter set, sharply reducing what each client must transmit.

Gap. Both FuzzyFed-CNN and the federated SAM-Med3D study use standard client participation, either near-uniform participation across simulated clients or standard random sampling, and neither explores how the choice of which clients to aggregate in a given round should depend on the quality or completeness of each client's local data.

2.1.3 Client Selection Strategies in Federated Learning

The general federated learning literature has studied client selection independently of any medical application. Li, Chen, and Teng [4] survey client selection strategies broadly, categorising approaches based on computational resources, data quality, and network conditions, and identify statistical heterogeneity across clients as a central challenge for both convergence speed and fairness. Tang et al. [12] propose FedCor, a correlation-based active client selection strategy that estimates which clients' updates are most likely to reduce the global loss and prioritises them, using a Gaussian-process model of inter-client loss correlation to guide selection without needing labelled validation data from each client.

Gap. These strategies are domain-agnostic. FedCor's correlation model is built from the statistics of model updates, not from any notion of medical data quality, modality completeness, or diagnostic class balance, so it does not by itself capture the constraints that are specific to a federation of hospitals holding multimodal Alzheimer's-diagnosis data.

2.1.4 Explainability in Medical AI

Because a clinician acting on a model's output needs to know why the model produced it, explainability is treated in this project as a first-class requirement rather than an afterthought. Selvaraju et al. [10] introduced Grad-CAM, which uses the gradient of a target class score with respect to a convolutional or attention feature map to produce a coarse localisation of the image regions that most influenced a prediction, and it remains a standard tool for explaining image-based medical predictions. Lundberg and Lee [11] introduced SHAP, a unified, game-theoretically grounded framework for attributing a model's prediction to its input features, which is directly applicable to the structured clinical and demographic variables used in this project. Mohanraj and Radhakrishnan [1] already demonstrate the value of an interpretable, fuzzy-inference layer for Alzheimer's diagnosis, reinforcing that explainability and diagnostic performance are not in conflict for this task.

Gap. None of the reviewed federated Alzheimer's-diagnosis systems pairs image-level explanation (Grad-CAM) with clinical-feature-level explanation (SHAP) within a single federated, multimodal pipeline, and none connects the explanation to which clients contributed most to a given prediction.

2.2 Problem Analysis

The four gaps identified above converge on a single structural problem. Table 2.1 contrasts the closest prior systems with the requirements of this project.

Table 2.1: Contrast between the closest prior work and the proposed framework.

System	**Modality**	**Client Selection**	**Foundation Model**	**Explainability**
FuzzyFed-CNN [1]	MRI (fuzzy fusion)	Near-uniform	No (CNN)	Fuzzy rules
ADLIP [2]	MRI + clinical text	N/A (centralised)	Yes (vision-language)	Limited
Federated SAM-Med3D [3]	MRI	Random/full	Yes (SAM-Med3D)	No
FedCor [12]	Generic (non-medical)	Correlation-based	No	No
Proposed	MRI + clinical	Adaptive, quality-aware	Yes (ViT-based)	Grad-CAM + SHAP

Three observations follow from Table 2.1. First, every prior system occupies a single cell of the modality-and-selection space: ADLIP has the multimodal foundation model but not the federated setting; FuzzyFed-CNN and the federated SAM-Med3D study have the federated, medical setting but only naive client participation; FedCor has principled client selection but neither the medical domain nor the multimodal foundation model. Second, none of the reviewed federated medical systems reports an explainability component that spans both imaging and clinical modalities. Third, no system in this comparison combines all three requirements, a federated multimodal vision foundation model, adaptive quality-aware client selection, and cross-modal explainability, in one framework.

The problem this project addresses is therefore how to train a single multimodal federated model that predicts Alzheimer's disease status from MRI and clinical data, using a client selection mechanism that accounts for data quality, modality completeness, and class balance instead of random or full participation, while keeping the model's decisions interpretable to clinicians through modality-specific explanations. Chapter 3 sets out the framework and methodology that respond to this problem.

Summary

This chapter reviewed the literature across four themes. Vision foundation models have proven effective for Alzheimer's diagnosis but remain largely centralised. Federated learning provides the privacy-preserving mechanism needed to train across hospitals, and has already been combined with Alzheimer's-specific multimodal models and with foundation-model fine-tuning, but always under naive client participation. General federated learning research has produced principled client selection strategies, but none tailored to medical multimodal data. Explainability tools exist for both imaging and structured clinical data individually, but have not been combined within a federated Alzheimer's-diagnosis pipeline. The gap that remains, and that Chapter 3 addresses, is a single federated, multimodal, foundation-model framework with adaptive client selection and cross-modal explainability.

Chapter 3

Research Methodology

This chapter describes the design of the proposed framework and the reasoning behind each design decision. Section 3.1 presents the framework as a whole, Section 3.2 describes the constituent datasets and their preparation, Section 3.3 gives the model definition and training algorithm, and Section 3.4 describes the individual modules and the planned ablation studies. The complete architecture is shown in Figure 3.1.

3.1 Proposed Framework

The framework accepts, at each client, an MRI scan and an accompanying record of clinical and demographic variables. Each client encodes its MRI data with a shared pretrained Vision Transformer (ViT) backbone and its clinical data with a lightweight feature encoder; the two representations are fused into a single client-level representation, which is fine-tuned locally using LoRA [7] so that only a small number of low-rank parameters are updated and transmitted. Before each aggregation round, every client computes a local score reflecting its data quality, modality completeness, and class balance. A central adaptive client selection module ranks clients by this score and selects the subset most likely to improve the global model. The central server then performs a weighted aggregation, analogous to FedAvg [9] but with client contributions weighted by their selection score, to produce an updated global model, which is broadcast back to all clients for the next round. After training, Grad-CAM [10] and SHAP [11] are applied to the imaging and clinical branches respectively to produce an explanation for each prediction.

Two design decisions in the framework are adopted from prior work and are credited as such rather than presented as contributions. Federated averaging of client updates is established by McMahan et al. [9], and parameter-efficient adaptation of a large pretrained model via low-rank updates is established by Hu et al. [7]. What is not established in prior work, and what this framework contributes, is the client scoring function and the weighted selection-and-aggregation procedure that uses it, combined with a multimodal vision foundation model fine-tuned under this scheme specifically for Alzheimer's disease prediction.

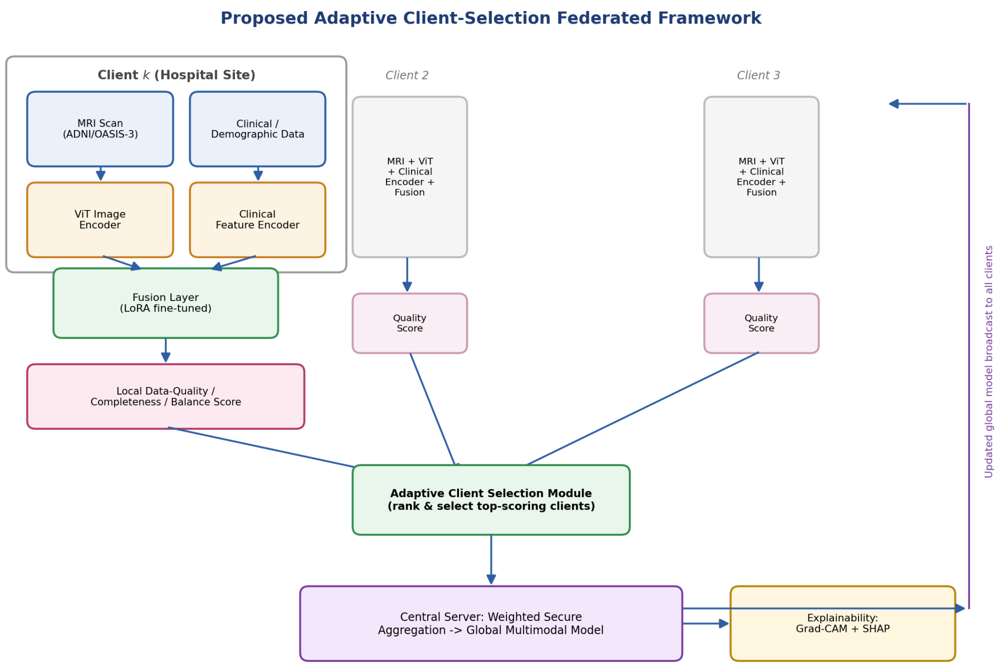

Figure 3.1: Proposed adaptive client-selection federated framework for multimodal Alzheimer's disease prediction.

3.2 Data Set Analysis / Software Project Requirement

3.2.1 Constituent Datasets

No single publicly available resource supplies both the imaging and clinical modalities this framework requires at a scale suitable for simulating a realistic hospital federation, so the framework is designed around two complementary sources.

The Alzheimer's Disease Neuroimaging Initiative (ADNI) [5] is a longitudinal, multi-site study that has enrolled participants across several phases (ADNI-1, ADNI-GO, ADNI-2, ADNI-3, and the ongoing ADNI-4), collectively comprising well over 2,000 participants across cognitively normal, mild cognitive impairment, and Alzheimer's disease groups, with structural MRI, PET imaging, genetic, and cognitive data collected at more than 60 research sites in the United States and Canada. OASIS-3 [6] is a retrospective, longitudinal compilation of data for 1,378 participants collected over three decades at the Washington University Knight Alzheimer Disease Research Center, comprising 755 cognitively normal adults and 622 individuals at varying stages of cognitive decline, with over 2,800 MRI sessions and more than 2,100 PET scans. Because both resources were collected at multiple research sites, they are well suited to simulating a non-IID federation: each site or acquisition cohort can be treated as a distinct client with its own scanner characteristics and population profile.

To supply the clinical and demographic modality at a scale and format suited to rapid experimentation, the framework additionally uses a structured Alzheimer's risk-factor dataset covering demographic variables (age, gender, country), lifestyle and cardiovascular risk factors, cognitive test scores, and genetic markers (including APOE $\varepsilon 4$ status) for a large number of individuals drawn from multiple countries. This tabular resource is used to represent the clinical branch of each client and, because it records a country field for every instance, provides a natural, realistic basis for partitioning instances into non-IID clients by country, as described in Section 3.2.2, without requiring an artificial partitioning scheme.

3.2.2 Non-IID Client Partitioning

Clients are simulated by partitioning the consolidated data along a site or country axis rather than uniformly at random, so that each client reflects a plausible single-institution population rather than an artificially balanced sample. For the imaging datasets, partitioning follows acquisition site or cohort; for the clinical dataset, partitioning follows the recorded country field. Under this scheme, clients differ in sample size, in the prevalence of each diagnostic class, and, where imaging and clinical records must be linked, in the completeness with which both modalities are available for a given individual. This is a deliberate design choice: a scheme that balanced classes and modality completeness across clients would remove exactly the heterogeneity that the adaptive client selection module is designed to address.

3.2.3 Preprocessing

MRI volumes are resampled to a common resolution and normalised to the intensity statistics expected by the pretrained ViT backbone, then divided into fixed-size patches for input to the transformer, following standard ViT preprocessing [8]. Clinical and demographic variables are normalised (numeric fields) or one-hot encoded (categorical fields), and missing values are flagged with an explicit indicator rather than imputed silently, so that the client-level completeness score described in Section 3.3.2 can be computed directly from these indicators.

3.3 Algorithm/Model Analysis

3.3.1 Model Definition

Each client k holds an MRI scan x_v and a clinical feature vector x_c (either may be partially missing). The imaging branch is encoded by a shared pretrained Vision Transformer,

$$h_v = \mathrm{ViT}(x_v) \in \mathbb{R}^d, \tag{3.1}$$

and the clinical branch by a lightweight feed-forward encoder,

$$h_c = \mathrm{MLP}(x_c) \in \mathbb{R}^d. \tag{3.2}$$

The two representations are fused by concatenation followed by a shared fully connected layer,

$$h = \mathrm{ReLU}(W_f[h_v; h_c] + b_f) \in \mathbb{R}^{d_s}, \tag{3.3}$$

and a classification head reads the diagnostic prediction from h,

$$p = \mathrm{softmax}(W_o h + b_o). \tag{3.4}$$

Only a low-rank update to the ViT backbone's weight matrices is trained locally at each client, following LoRA [7]: for a pretrained weight matrix $W_0 \in \mathbb{R}^{d\times d}$, the effective local weight is

$$W = W_0 + \Delta W, \qquad \Delta W = BA, \quad B \in \mathbb{R}^{d\times r},\ A \in \mathbb{R}^{r\times d},\ r \ll d, \tag{3.5}$$

so that only A and B, not W_0, are updated and transmitted, sharply reducing the size of each client's communication payload.

3.3.2 Adaptive Client Selection Scoring

Before round t, each client k computes a score combining three signals: local data quality Q_k (e.g. image resolution/artifact indicators and the fraction of non-missing clinical fields), modality completeness C_k (the fraction of local instances with both an MRI scan and a linked clinical record), and class balance B_k (the inverse of the class-imbalance ratio within the client's local diagnostic labels). The client score is a weighted combination,

$$S_k = w_Q Q_k + w_C C_k + w_B B_k, \qquad w_Q + w_C + w_B = 1, \tag{3.6}$$

with the weights either fixed by validation or learned jointly with the model. The server selects the top-m clients by score, $\mathcal{K}_t = \mathrm{TopM}(S_1, \ldots, S_K)$, for participation in round t.

3.3.3 Federated Aggregation

Selected clients' local updates are aggregated with weights proportional to both local dataset size n_k and selection score S_k, extending FedAvg [9]:

$$\theta_{t+1} = \sum_{k \in \mathcal{K}_t} \frac{n_k S_k}{\sum_{j \in \mathcal{K}_t} n_j S_j} \, \theta_k^{(t)}. \tag{3.7}$$

Algorithm 1 states the client selection procedure and Algorithm 2 states one federated training round.

Algorithm 1 Adaptive client selection

Require: client set $\{1, \ldots, K\}$, weights w_Q, w_C, w_B, round size m
Ensure: selected clients $\mathcal{K}_t$

```
for each client k do
    compute Q_k, C_k, B_k from local data
    S_k ← w_Q Q_k + w_C C_k + w_B B_k
end for
K_t ← indices of the m largest scores among {S_1, ..., S_K}
return K_t
```

Algorithm 2 One federated training round

Require: global parameters θ_t, selected clients $\mathcal{K}_t$
Ensure: updated global parameters θ_{t+1}

```
for each client k ∈ K_t (in parallel) do
    initialise local LoRA parameters from θ_t
    fine-tune locally on (x_v, x_c, y) using Eqs. (1)–(4)
    return updated local parameters θ_k^(t) and score S_k
end for
aggregate: θ_{t+1} ← Eq. (7) over K_t
return θ_{t+1}
```

3.4 Explanation of Different Modules

3.4.1 Vision Foundation Model Encoder

The imaging branch uses a Vision Transformer backbone pretrained on a large natural- or medical-image corpus, adapted to MRI slices through the LoRA-based fine-tuning described in Section 3.3.1. Keeping the backbone weights frozen except for the low-rank update follows the finding of Mouheb et al. [3] that freezing the encoder of a medical foundation model can match full fine-tuning while substantially reducing training and communication cost.

3.4.2 Clinical Feature Encoder

The clinical branch encodes demographic, lifestyle, cognitive, and genetic risk variables through a small feed-forward network, which is deliberately kept lightweight relative to the imaging branch since the clinical modality has far fewer input dimensions and does not require a pretrained backbone.

3.4.3 Client Selection Module

The client selection module is the central contribution of this project. It is implemented as a lightweight scoring function (Section 3.3.2) computed locally at each client from statistics that do not require exposing raw patient data, so that the privacy guarantees of the federated setting are preserved even though client quality is now taken into account.

3.4.4 Explainability Module

Grad-CAM [10] is applied to the ViT encoder's attention maps to highlight the image regions contributing most to a prediction, and SHAP [11] is applied to the clinical encoder's input features to attribute the prediction to specific clinical or demographic variables. The two attributions are presented together so that a clinician reviewing a flagged case sees both the imaging evidence and the clinical evidence behind it.

3.4.5 Planned Ablations

Four ablations are planned. The first and central ablation replaces adaptive client selection with random sampling, isolating the effect of the selection mechanism itself. The second replaces the weighted aggregation of Equation (7) with plain, unweighted FedAvg [9]. The third restricts the input to a single modality (MRI-only or clinical-only) to measure the contribution of multimodal fusion. The fourth compares the proposed scoring function against FedCor's correlation-based selection [12] to determine whether a medical-domain-aware score offers a measurable advantage over a domain-agnostic one.

Summary

This chapter has set out the proposed framework and the reasoning behind it. A shared ViT-based imaging encoder and a lightweight clinical encoder are fused and fine-tuned locally using LoRA, clients are scored on data quality, modality completeness, and class balance before each round, and the server aggregates the selected clients' updates with weights proportional to both dataset size and selection score. ADNI and OASIS-3 supply the imaging modality and a structured risk-factor dataset supplies the clinical modality, partitioned by site or country to produce a realistic non-IID federation. Chapter 4 will report the implementation of this design and its evaluation against the FedAvg and FedCor baselines once experiments are conducted.

Chapter 4

Implementation and Result Analysis

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4.1 Environment Setup

Write Environment Setup in chapter4.tex

4.2 Data Preprocessing

Write Data Preprocessing steps here in chapter4.tex

4.3 Model Implementation

Write on Model Implementation here in chapter4.tex

4.4 Evaluation Metrics

Write Evaluation Metrics in chapter4.tex

4.5 Performance Analysis

Write Performance Analysis in chapter4.tex

4.6 Key Findings and Discussions

Write Key Findings and Discussion in chapter4.tex

4.7 Comparison with State of the Art Model

Write a Comparison with the State of the Art Model in chapter4.tex

Summary

Write a summary passage on this chapter here in chapter4.tex

Chapter 5

Standards, Constraints, and Milestones

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5.1 Sustainability Standards

Write Standards in chapter5.tex

5.2 Impacts on Society

Write Impacts on Society in chapter5.tex

5.3 Ethics

Write Ethics in chapter5.tex

5.4 Challenges

Write Challenges in chapter5.tex

5.5 Constraints

Write different Constraints like Design constraints, Component Constraints, and Budget constraints in chapter5.tex

5.6 Timeline and Gantt Chart

Write Timeline and Gantt Chart in chapter5.tex

Summary

Write a summary passage of Standards, Constraints, and Milestones at the end of chapter5.tex

Chapter 6

Conclusion

Write the conclusion passage in chapter6.tex

6.1 Limitations

Write about the limitations of your work in chapter6.tex

6.2 Future Works and Direction

Write Future Works and Direction in chapter6.tex

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